

CIRCLE COMPACTIFICATION AND T-DUALITY

STRING THEORY EXAM

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MOTIVATION

Superstring lives in 10d.

Superstring lives in 10d.
We live in 4d.

Superstring lives in 10d.

We live in 4d.

What to do with those extra-dimensions?

WHAT REALLY IS SPACETIME?

KALUZA-KLEIN COMPACTIFICATION

KALUZA-KLEIN COMPACTIFICATION: MAIN RESULTS

Compactification of a **scalar field** on $\mathcal{M}_D \rightarrow \mathcal{M}_{D-1} \times S^1$:

Kaluza-Klein Masses

$$m_s^2 = M^2 + \frac{s^2}{R^2}.$$

Negligible for

$$E \ll \frac{1}{R}.$$

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Compactification of **Einstein gravity** on $\mathcal{M}_D \rightarrow \mathcal{M}_{D-1} \times S^1$:

Result

- Decomposition $SO(1, D-1) \rightarrow SO(1, D-2)$;
- Gauge symmetry;
- $\text{Vol}(S^1) = \int_0^{2\pi R} dy \sqrt{G_{yy}^{(0)}} = e^{-(D-3)\alpha_{D-1}\phi} \cdot 2\pi R.$

SUPERSTRING

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- Critical setting $\mathcal{M}_{10}$. Closed string;
- Bosonic X^μ and fermionic ψ^μ fields;

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- $X^i = X_L^i(\tau + \sigma) + X_R^i(\tau, \sigma)$, $\psi^i = \psi_L^i(\tau + \sigma) + \psi_R^i(\tau - \sigma)$;

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- Mode expansion: $\psi_{L/R}^i(\xi^\pm) = \sqrt{\frac{2\pi}{l}} \sum_{r \in \mathbb{Z} + \phi} b_r^{i(\sim)} e^{-\frac{2\pi i}{l} r \xi^\pm}$;

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- NS sector: $\phi = 1/2$, $a^{(\sim)} = 1/2$;
- R sector: $\phi = 0$, $a^{(\sim)} = 0$;

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- Mode expansion: $\psi_{L/R}^i(\xi^\pm) = \sqrt{\frac{2\pi}{l}} \sum_{r \in \mathbb{Z} + \phi} b_r^{(\sim)i} e^{-\frac{2\pi i}{l} r \xi^\pm}$;
- NS sector: $\phi = 1/2$, $\tilde{a} = 1/2$;
- R sector: $\phi = 0$, $\tilde{a} = 0$;
- Level-matching condition: $M_L^2 = M_R^2$;
- Mass-shell condition: $M_{L/R}^2 = \frac{2}{\alpha'} \left(N_\perp - \tilde{a} \right)$.

SUPERSTRING: THE SECTORS

sector	G-parity	state	little group rep.	$\alpha' M_R^2/2$	statistics
NS	—	$ 0\rangle_{NS}$	$SO(9) : \mathbf{1}$	$-1/2$	boson
NS	+	$b_{-1/2}^i 0\rangle_{NS}$	$SO(8) : \mathbf{8}_v$	0	boson
R	+	$ 0\rangle_R, B_{a_1}^+ B_{a_2}^+ 0\rangle_R,$ $B_1^+ B_2^+ B_3^+ B_4^+ 0\rangle_R$	$SO(8) : \mathbf{8}_s$	0	fermion
R	—	$B_{a_1}^+ 0\rangle_R,$ $B_{a_1}^+ B_{a_2}^+ B_{a_3}^+ 0\rangle_R$	$SO(8) : \mathbf{8}_c$	0	fermion

SUPERSTRING: TYPE II THEORIES

Type IIA			Type IIB		
sector	fields	$SO(8)$	sector	fields	$SO(8)$
(NS_+, NS_+)	$\Phi, B_{[\mu\nu]}, G_{(\mu\nu)}$	$\mathbf{8}_v \otimes \mathbf{8}_v$	(NS_+, NS_+)	$\Phi, B_{[\mu\nu]}, G_{(\mu\nu)}$	$\mathbf{8}_v \otimes \mathbf{8}_v$
(R_+, R_-)	C_1, C_3	$\mathbf{8}_s \otimes \mathbf{8}_c$	(R_+, R_+)	C_0, C_2, C_4^+	$\mathbf{8}_s \otimes \mathbf{8}_s$
(NS_+, R_-)	$\tilde{\lambda}_a, \tilde{\psi}_a^\mu$	$\mathbf{8}_v \otimes \mathbf{8}_c$	(NS_+, R_+)	$\lambda_a^{(1)}, \psi_a^{(1)\mu}$	$\mathbf{8}_v \otimes \mathbf{8}_s$
(R_+, NS_+)	λ_a, ψ_a^μ	$\mathbf{8}_s \otimes \mathbf{8}_v$	(R_+, NS_+)	$\lambda_a^{(2)}, \psi_a^{(2)\mu}$	$\mathbf{8}_s \otimes \mathbf{8}_v$

TYPE II STRING: COMPACTIFICATION

TYPE II STRING: FIELD THEORETIC COMPACTIFICATION

Type IIA					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_-)	$\mathbf{8}_s \otimes \mathbf{8}_c$	$C_{\hat{\mu}}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}]}$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$A_9, A_\mu, C_{9\mu\nu}, C_{\mu\nu\rho}$
(NS_+, R_-)	$\mathbf{8}_v \otimes \mathbf{8}_c$	$\tilde{\lambda}_a, \tilde{\psi}_a^{\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\tilde{\psi}_a^\mu, \tilde{\psi}_a^9$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\tilde{\lambda}_a$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a, \psi_a^{\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	ψ_a^μ, ψ_a^9
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	λ_a
Type IIB					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_+)	$\mathbf{8}_s \otimes \mathbf{8}_s$	$C_0, C_{[\hat{\mu}\hat{\nu}]}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}]}^+$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$a, C_{\mu 9}, C_{\mu\nu}, C_{\mu\nu\rho 9}$
(NS_+, R_+)	$\mathbf{8}_v \otimes \mathbf{8}_s$	$\lambda_a^{(1)}, \psi_a^{(1)\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(1)9}, \psi_a^{(1)\mu}$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\lambda_a^{(1)}$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a^{(2)}, \psi_a^{(2)\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(2)9}, \psi_a^{(2)\mu}$
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	$\lambda_a^{(2)}$

TYPE II STRING: FIELD THEORETIC COMPACTIFICATION

Type IIA					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
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(R_+, R_-)	$\mathbf{8}_s \otimes \mathbf{8}_c$	$C_{\hat{\mu}}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}]}$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$A_9, A_\mu, C_{9\mu\nu}, C_{\mu\nu\rho}$
(NS_+, R_-)	$\mathbf{8}_v \otimes \mathbf{8}_c$	$\tilde{\lambda}_a, \tilde{\psi}_a^{\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\tilde{\psi}_a^\mu, \tilde{\psi}_a^9$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\tilde{\lambda}_a$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a, \psi_a^{\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	ψ_a^μ, ψ_a^9
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	λ_a
Type IIB					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_+)	$\mathbf{8}_s \otimes \mathbf{8}_s$	$C_0, C_{[\hat{\mu}\hat{\nu}]}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}]}^+$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$a, C_{\mu 9}, C_{\mu\nu}, C_{\mu\nu\rho 9}$
(NS_+, R_+)	$\mathbf{8}_v \otimes \mathbf{8}_s$	$\lambda_a^{(1)}, \psi_a^{(1)\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(1)9}, \psi_a^{(1)\mu}$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\lambda_a^{(1)}$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a^{(2)}, \psi_a^{(2)\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(2)9}, \psi_a^{(2)\mu}$
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sector	$SO(8)$	10d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$
(R_+, R_-)	$\mathbf{8}_s \otimes \mathbf{8}_c$	$C_{\hat{\mu}}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}]}$
(NS_+, R_-)	$\mathbf{8}_v \otimes \mathbf{8}_c$	$\tilde{\lambda}_a, \tilde{\psi}_a^{\hat{\mu}}$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a, \psi_a^{\hat{\mu}}$

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sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
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			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
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(NS_+, R_-)	$\mathbf{8}_v \otimes \mathbf{8}_c$	$\tilde{\lambda}_a, \tilde{\psi}_a^{\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\tilde{\psi}_a^\mu, \tilde{\psi}_a^9$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\tilde{\lambda}_a$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a, \psi_a^{\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	ψ_a^μ, ψ_a^9
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	λ_a
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sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_+)	$\mathbf{8}_s \otimes \mathbf{8}_s$	$C_0, C_{[\hat{\mu}\hat{\nu}]}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}]}^+$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$a, C_{\mu 9}, C_{\mu\nu}, C_{\mu\nu\rho 9}$
(NS_+, R_+)	$\mathbf{8}_v \otimes \mathbf{8}_s$	$\lambda_a^{(1)}, \psi_a^{(1)\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(1)9}, \psi_a^{(1)\mu}$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\lambda_a^{(1)}$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a^{(2)}, \psi_a^{(2)\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(2)9}, \psi_a^{(2)\mu}$
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TYPE II STRING: FIELD THEORETIC COMPACTIFICATION

Type IIA					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_-)	$\mathbf{8}_s \otimes \mathbf{8}_c$	$C_{\hat{\mu}}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}]}$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$A_9, A_\mu, C_{9\mu\nu}, C_{\mu\nu\rho}$
(NS_+, R_-)	$\mathbf{8}_v \otimes \mathbf{8}_c$	$\tilde{\lambda}_a, \tilde{\psi}_a^{\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\tilde{\psi}_a^\mu, \tilde{\psi}_a^9$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\tilde{\lambda}_a$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a, \psi_a^{\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	ψ_a^μ, ψ_a^9
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	λ_a
Type IIB					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_+)	$\mathbf{8}_s \otimes \mathbf{8}_s$	$C_0, C_{[\hat{\mu}\hat{\nu}]}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}]}^+$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$a, C_{\mu 9}, C_{\mu\nu}, C_{\mu\nu\rho 9}$
(NS_+, R_+)	$\mathbf{8}_v \otimes \mathbf{8}_s$	$\lambda_a^{(1)}, \psi_a^{(1)\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(1)9}, \psi_a^{(1)\mu}$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\lambda_a^{(1)}$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a^{(2)}, \psi_a^{(2)\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(2)9}, \psi_a^{(2)\mu}$
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	$\lambda_a^{(2)}$

Type IIA

$SO(7)$	$SO(7)$ irrep	9d fields
$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$A_9, A_\mu, C_{9\mu\nu}, C_{\mu\nu\rho}$
$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\tilde{\psi}_a^\mu, \tilde{\psi}_a^9$
$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\tilde{\lambda}_a$
$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	ψ_a^μ, ψ_a^9
$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	λ_a

TYPE II STRING: FIELD THEORETIC COMPACTIFICATION

Type IIA					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_-)	$\mathbf{8}_s \otimes \mathbf{8}_c$	$C_{\hat{\mu}}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}]}$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$A_9, A_\mu, C_{9\mu\nu}, C_{\mu\nu\rho}$
(NS_+, R_-)	$\mathbf{8}_v \otimes \mathbf{8}_c$	$\tilde{\lambda}_a, \tilde{\psi}_a^{\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\tilde{\psi}_a^\mu, \tilde{\psi}_a^9$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\tilde{\lambda}_a$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a, \psi_a^{\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	ψ_a^μ, ψ_a^9
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	λ_a
Type IIB					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_+)	$\mathbf{8}_s \otimes \mathbf{8}_s$	$C_0, C_{[\hat{\mu}\hat{\nu}]}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}]}^+$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$a, C_{\mu 9}, C_{\mu\nu}, C_{\mu\nu\rho 9}$
(NS_+, R_+)	$\mathbf{8}_v \otimes \mathbf{8}_s$	$\lambda_a^{(1)}, \psi_a^{(1)\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(1)9}, \psi_a^{(1)\mu}$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\lambda_a^{(1)}$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a^{(2)}, \psi_a^{(2)\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(2)9}, \psi_a^{(2)\mu}$
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	$\lambda_a^{(2)}$

TYPE II STRING: FIELD THEORETIC COMPACTIFICATION

Type IIA					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
$(\text{NS}_+, \text{NS}_+)$	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_-)	$\mathbf{8}_s \otimes \mathbf{8}_c$	$C_{\hat{\mu}}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}]}$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$A_9, A_\mu, C_{9\mu\nu}, C_{\mu\nu\rho}$
$(\text{NS}_+, \text{R}_-)$	$\mathbf{8}_v \otimes \mathbf{8}_c$	$\tilde{\lambda}_a, \tilde{\psi}_a^{\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\tilde{\psi}_a^\mu, \tilde{\psi}_a^9$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\tilde{\lambda}_a$
$(\text{R}_+, \text{NS}_+)$	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a, \psi_a^{\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	ψ_a^μ, ψ_a^9
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	λ_a
Type IIB					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
$(\text{NS}_+, \text{NS}_+)$	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_+)	$\mathbf{8}_s \otimes \mathbf{8}_s$	$C_0, C_{[\hat{\mu}\hat{\nu}]}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}]}^+$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$a, C_{\mu 9}, C_{\mu\nu}, C_{\mu\nu\rho 9}$
$(\text{NS}_+, \text{R}_+)$	$\mathbf{8}_v \otimes \mathbf{8}_s$	$\lambda_a^{(1)}, \psi_a^{(1)\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(1)9}, \psi_a^{(1)\mu}$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\lambda_a^{(1)}$
$(\text{R}_+, \text{NS}_+)$	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a^{(2)}, \psi_a^{(2)\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(2)9}, \psi_a^{(2)\mu}$
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	$\lambda_a^{(2)}$

sector	$SO(8)$	10d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$
(R_+, R_+)	$\mathbf{8}_s \otimes \mathbf{8}_s$	$C_0, C_{[\hat{\mu}\hat{\nu}]}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}]}^+$
(NS_+, R_+)	$\mathbf{8}_v \otimes \mathbf{8}_s$	$\lambda_a^{(1)}, \psi_a^{(1)\hat{\mu}}$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a^{(2)}, \psi_a^{(2)\hat{\mu}}$

TYPE II STRING: FIELD THEORETIC COMPACTIFICATION

Type IIA					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_-)	$\mathbf{8}_s \otimes \mathbf{8}_c$	$C_{\hat{\mu}}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}]}$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$A_9, A_\mu, C_{9\mu\nu}, C_{\mu\nu\rho}$
(NS_+, R_-)	$\mathbf{8}_v \otimes \mathbf{8}_c$	$\tilde{\lambda}_a, \tilde{\psi}_a^{\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\tilde{\psi}_a^\mu, \tilde{\psi}_a^9$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\tilde{\lambda}_a$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a, \psi_a^{\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	ψ_a^μ, ψ_a^9
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	λ_a
Type IIB					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_+)	$\mathbf{8}_s \otimes \mathbf{8}_s$	$C_0, C_{[\hat{\mu}\hat{\nu}]}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}]}^+$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$a, C_{\mu 9}, C_{\mu\nu}, C_{\mu\nu\rho 9}$
(NS_+, R_+)	$\mathbf{8}_v \otimes \mathbf{8}_s$	$\lambda_a^{(1)}, \psi_a^{(1)\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(1)9}, \psi_a^{(1)\mu}$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\lambda_a^{(1)}$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a^{(2)}, \psi_a^{(2)\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(2)9}, \psi_a^{(2)\mu}$
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	$\lambda_a^{(2)}$

TYPE II STRING: FIELD THEORETIC COMPACTIFICATION

Type IIA					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_-)	$\mathbf{8}_s \otimes \mathbf{8}_c$	$C_{\hat{\mu}}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}]}$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$A_9, A_\mu, C_{9\mu\nu}, C_{\mu\nu\rho}$
(NS_+, R_-)	$\mathbf{8}_v \otimes \mathbf{8}_c$	$\tilde{\lambda}_a, \tilde{\psi}_a^{\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\tilde{\psi}_a^\mu, \tilde{\psi}_a^9$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\tilde{\lambda}_a$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a, \psi_a^{\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	ψ_a^μ, ψ_a^9
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	λ_a
Type IIB					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
(NS_+, NS_+)	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_+)	$\mathbf{8}_s \otimes \mathbf{8}_s$	$C_0, C_{[\hat{\mu}\hat{\nu}]}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}]}^+$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$a, C_{\mu 9}, C_{\mu\nu}, C_{\mu\nu\rho 9}$
(NS_+, R_+)	$\mathbf{8}_v \otimes \mathbf{8}_s$	$\lambda_a^{(1)}, \psi_a^{(1)\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(1)9}, \psi_a^{(1)\mu}$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\lambda_a^{(1)}$
(R_+, NS_+)	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a^{(2)}, \psi_a^{(2)\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(2)9}, \psi_a^{(2)\mu}$
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	$\lambda_a^{(2)}$

Type IIB

$SO(7)$	$SO(7)$ irrep	9d fields
$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$a, C_{\mu 9}, C_{\mu\nu}, C_{\mu\nu\rho 9}$
$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(1)9}, \psi_a^{(1)\mu}$
$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\lambda_a^{(1)}$
$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(2)9}, \psi_a^{(2)\mu}$
$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	$\lambda_a^{(2)}$

TYPE II STRING: FIELD THEORETIC COMPACTIFICATION

Type IIA					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
$(\text{NS}_+, \text{NS}_+)$	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_-)	$\mathbf{8}_s \otimes \mathbf{8}_c$	$C_{\hat{\mu}}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}]}$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$A_9, A_\mu, C_{9\mu\nu}, C_{\mu\nu\rho}$
$(\text{NS}_+, \text{R}_-)$	$\mathbf{8}_v \otimes \mathbf{8}_c$	$\tilde{\lambda}_a, \tilde{\psi}_a^{\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\tilde{\psi}_a^\mu, \tilde{\psi}_a^9$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\tilde{\lambda}_a$
$(\text{R}_+, \text{NS}_+)$	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a, \psi_a^{\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	ψ_a^μ, ψ_a^9
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	λ_a
Type IIB					
sector	$SO(8)$	10d fields	$SO(7)$	$SO(7)$ irrep	9d fields
$(\text{NS}_+, \text{NS}_+)$	$\mathbf{8}_v \otimes \mathbf{8}_v$	$\Phi, B_{[\hat{\mu}\hat{\nu}]}, G_{(\hat{\mu}\hat{\nu})}$	$\mathbf{7} \otimes \mathbf{7}$	$\mathbf{1} \oplus \mathbf{21} \oplus \mathbf{27}$	$\phi, B_{[\mu\nu]}, G_{(\mu\nu)}$
			$(\mathbf{7} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{7})$	$\mathbf{7} \oplus \mathbf{7}$	$G_{\mu 9}, B_{\mu 9}$
			$\mathbf{1} \otimes \mathbf{1}$	$\mathbf{1}$	G_{99}
(R_+, R_+)	$\mathbf{8}_s \otimes \mathbf{8}_s$	$C_0, C_{[\hat{\mu}\hat{\nu}]}, C_{[\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}]}^+$	$\mathbf{8} \otimes \mathbf{8}$	$\mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}$	$a, C_{\mu 9}, C_{\mu\nu}, C_{\mu\nu\rho 9}$
$(\text{NS}_+, \text{R}_+)$	$\mathbf{8}_v \otimes \mathbf{8}_s$	$\lambda_a^{(1)}, \psi_a^{(1)\hat{\mu}}$	$\mathbf{7} \otimes \mathbf{8}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(1)9}, \psi_a^{(1)\mu}$
			$\mathbf{1} \otimes \mathbf{8}$	$\mathbf{8}$	$\lambda_a^{(1)}$
$(\text{R}_+, \text{NS}_+)$	$\mathbf{8}_s \otimes \mathbf{8}_v$	$\lambda_a^{(2)}, \psi_a^{(2)\hat{\mu}}$	$\mathbf{8} \otimes \mathbf{7}$	$\mathbf{8} \oplus \mathbf{48}$	$\psi_a^{(2)9}, \psi_a^{(2)\mu}$
			$\mathbf{8} \otimes \mathbf{1}$	$\mathbf{8}$	$\lambda_a^{(2)}$

IS THIS A COINCIDENCE?

IS THIS A COINCIDENCE?
IS THIS A FEATURE OF THE MASSLESS SPECTRUM?

Setting and Main Results

Lightcone quantization:

- $\mathcal{M}_{10} \rightarrow \mathcal{M}_9 \times S^1, X^9 \simeq X^9 + 2\pi R;$
- $X^i(\tau, \sigma + l) = X^i(\tau, \sigma);$

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- $X^i(\tau, \sigma + l) = X^i(\tau, \sigma);$
- $X^9(\tau, \sigma + l) = X^9(\tau, \sigma) + 2\pi R\omega;$
- $p_9 = s/R, s \in \mathbb{Z}.$

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- $p_9 = s/R, s \in \mathbb{Z}.$

Mode decomposition:

- $X_{L/R}^9(\xi^\pm) = \frac{x^9}{2} + \frac{\alpha'\pi}{l} p_{L/R} \xi^\pm + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{\alpha_n^9}{n} e^{-\frac{2\pi i}{l} n \xi^\pm};$
- $p_L = \left(\frac{s}{R} + \frac{\omega R}{\alpha'}\right), p_R = \left(\frac{s}{R} - \frac{\omega R}{\alpha'}\right);$

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- $\mathcal{M}_{10} \rightarrow \mathcal{M}_9 \times S^1, X^9 \simeq X^9 + 2\pi R;$
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- $X_{L/R}^9(\xi^\pm) = \frac{x^9}{2} + \frac{\alpha'\pi}{l} p_{L/R} \xi^\pm + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{\alpha_n^9}{n} e^{-\frac{2\pi i}{l} n \xi^\pm};$
- $p_L = \left(\frac{s}{R} + \frac{\omega R}{\alpha'}\right), p_R = \left(\frac{s}{R} - \frac{\omega R}{\alpha'}\right);$
- $M_L^2 = \frac{p_L^2}{2} + \frac{2}{\alpha'} \left(\tilde{N}_\perp - \tilde{a}\right), \quad M_R^2 = \frac{p_R^2}{2} + \frac{2}{\alpha'} (N_\perp - a);$
- $M_L^2 = M_R^2, M^2 = M_L^2 + M_R^2.$

T-DUALITY: INTRODUCTION

Mass-Shell Condition

$$M^2 = \frac{s^2}{R^2} + \frac{\omega^2 R^2}{\alpha'^2} + \frac{2}{\alpha'}(\tilde{N}_\perp + N_\perp - a - \tilde{a}),$$
$$p_L = \left(\frac{s}{R} + \frac{\omega R}{\alpha'} \right), \quad p_R = \left(\frac{s}{R} - \frac{\omega R}{\alpha'} \right).$$

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$$p_L = \left(\frac{s}{R} + \frac{\omega R}{\alpha'} \right), \quad p_R = \left(\frac{s}{R} - \frac{\omega R}{\alpha'} \right).$$

From Spectrum

$$R \rightarrow R' = \alpha'/R,$$
$$(s, \omega) \rightarrow (s', \omega') = (\omega, s).$$

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$$p_L = \left(\frac{s}{R} + \frac{\omega R}{\alpha'} \right), \quad p_R = \left(\frac{s}{R} - \frac{\omega R}{\alpha'} \right).$$

From Spectrum

$$R \rightarrow R' = \alpha'/R,$$
$$(s, \omega) \rightarrow (s', \omega') = (\omega, s).$$

From Full Theory

$$p_L \rightarrow p_L, \quad p_R \rightarrow -p_R,$$
$$X_L^9 \rightarrow X_L^9, \quad X_R^9 \rightarrow -X_R^9.$$

Mass-Shell Condition

$$M^2 = \frac{s^2}{R^2} + \frac{\omega^2 R^2}{\alpha'^2} + \frac{2}{\alpha'}(\tilde{N}_\perp + N_\perp - a - \tilde{a}),$$
$$p_L = \left(\frac{s}{R} + \frac{\omega R}{\alpha'} \right), \quad p_R = \left(\frac{s}{R} - \frac{\omega R}{\alpha'} \right).$$

From Spectrum

$$R \rightarrow R' = \alpha'/R,$$
$$(s, \omega) \rightarrow (s', \omega') = (\omega, s).$$

From Full Theory

$$p_L \rightarrow p_L, \quad p_R \rightarrow -p_R,$$
$$X_L^9 \rightarrow X_L^9, \quad X_R^9 \rightarrow -X_R^9.$$

from Worldsheet Supersymmetry

$$\psi_L^9 \rightarrow \psi_L^9, \quad \psi_R^9 \rightarrow -\psi_R^9.$$

T-Duality

Transformation:

$$\begin{aligned} X^9(\tau, \sigma) &= X_L^9(\xi^+) + X_R^9(\xi^-) \rightarrow X'^9(\tau, \sigma) = X_L^9(\xi^+) - X_R^9(\xi^-), \\ \psi^9(\tau, \sigma) &= \psi_L^9(\xi^+) + \psi_R^9(\xi^-) \rightarrow \psi'^9(\tau, \sigma) = \psi_L^9(\xi^+) - \psi_R^9(\xi^-). \end{aligned}$$

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On the modes:

$$\begin{aligned}\tilde{b}_r^9 &\rightarrow \tilde{b}'^9_r = \tilde{b}_r^9, & b_r^9 &\rightarrow b'^9_r = -b_r^9, \\ B_4^\pm &= \frac{1}{\sqrt{2}} (b_0^8 \pm i b_0^9) \rightarrow \frac{1}{\sqrt{2}} (b_0^8 \mp i b_0^9) = B_4^\mp.\end{aligned}$$

T-Duality

Transformation:

$$\begin{aligned} X^9(\tau, \sigma) &= X_L^9(\xi^+) + X_R^9(\xi^-) \rightarrow X'^9(\tau, \sigma) = X_L^9(\xi^+) - X_R^9(\xi^-), \\ \psi^9(\tau, \sigma) &= \psi_L^9(\xi^+) + \psi_R^9(\xi^-) \rightarrow \psi'^9(\tau, \sigma) = \psi_L^9(\xi^+) - \psi_R^9(\xi^-). \end{aligned}$$

On the modes:

$$\begin{aligned} \tilde{b}_r^9 &\rightarrow \tilde{b}'^9_r = \tilde{b}_r^9, & b_r^9 &\rightarrow b'^9_r = -b_r^9, \\ B_4^\pm &= \frac{1}{\sqrt{2}} (b_0^8 \pm i b_0^9) \rightarrow \frac{1}{\sqrt{2}} (b_0^8 \mp i b_0^9) = B_4^\mp. \end{aligned}$$

On the sectors:

$$\begin{aligned} (R_+, R_\pm) &\rightarrow (R_+, R_\mp), \\ (NS_+, R_\pm) &\rightarrow (NS_+, R_\mp). \end{aligned}$$

Type IIB on S^1 with $R \cong$ Type IIA on S'^1 with $R' = \frac{\alpha'}{R}$.

$$\begin{aligned}X^9(\tau, \sigma) &= X_L^9(\xi^+) + X_R^9(\xi^-), \\X'^9(\tau, \sigma) &= X_L^9(\xi^+) - X_R^9(\xi^-).\end{aligned}$$

THEN, WHAT REALLY IS SPACETIME?